SiPM and ASIC response: first pilot

NAUSICAA

The Julia response starts from saved incident photons, applies binomial PDE, uniform microcell illumination and exponential recovery, and adds Poisson primary dark pulses. The independent ASIC stage adds Gaussian PE noise and clips the bucket-average current at the configured upper limit. All avalanche charge is assigned to the trigger bucket. Crosstalk, afterpulsing and gain spread are deferred. Dark initial ages are stationary; the PDE is constant during recovery. Float64 charge accumulators preserve the sampled sensor charge. The physical model and source discussion are in `docs/sipm_electronics_model.tex`.

1 Saved parameters

Parameter	Value	Unit
PDE	0.3	
Effective microcells	7854	
Gain	2550000	electrons/PE
Recovery time	7.34e-08	s
Sensor diameter	0.005	m
Primary dark rate	2181662	Hz
Charge per fully recovered PE	4.08555e-13	C
Bucket duration	2.5e-07	s
ASIC noise RMS	0.5	PE
ASIC noise RMS	8.171101e-07	A
Upper average-current limit	0.001	A
Equivalent upper range	611.9127	PE/bucket
Mean primary dark pulses	0.5454154	per bucket

2 Saved pilot and validation results

The pilot uses event 0 in each of the four existing 250 ns photon histories. Channels illuminated anywhere in the event are acquired through the global photon-bucket span with four quiet buckets at either end. Zero-photon buckets contain dark pulses and ASIC noise. Recovery loss includes optical and dark charge. Clipping counts include every acquired channel–bucket cell. These four events establish operation of the pipeline; they do not estimate the full-sample saturation fraction.

Sample	Gas	Event	Detected PE	Recovery loss	Clipped buckets
bb0nu	pure_xe	0	13,782,681	0.12357	5,038 / 44,652
bb0nu	xe_he15	0	13,558,437	0.28869	2,360 / 18,550
single_electrons	pure_xe	0	14,493,517	0.23641	3,151 / 18,850
single_electrons	xe_he15	0	14,255,194	0.39132	1,552 / 9,088

bb0nu, pure_xe: maximum sensor charge 11697.641 PE; dark pulses 24,713; ASIC noise RMS 0.49769 PE.

bb0nu, xe_he15: maximum sensor charge 19665.466 PE; dark pulses 10,135; ASIC noise RMS 0.50167 PE.

single_electrons, pure_xe: maximum sensor charge 16097.155 PE; dark pulses 10,425; ASIC noise RMS 0.50524 PE.

single_electrons, xe_he15: maximum sensor charge 20557.517 PE; dark pulses 5,039; ASIC noise RMS 0.49691 PE.

2.1 Independent numerical checks

The two-trigger expectation is independent midpoint quadrature of the uniform-arrival separation density. The stationary test uses independent Poisson photon counts, 40 microcells and 80 detected triggers per bucket; the transient test begins with recovered cells. The inverse means demonstrate finite-bin bias and the difference between stationary and transient illumination. Dark-count tests use the 5 mm effective sensor; binomial tests use 100

incident photons at PDE 0.30. ASIC tests include variance, lag-one covariance, charge/current units and exact range boundaries. Interface checks include signed gaps, empty events, HDF5 round trips, seed replay and reuse of the sensor output.

Validation quantity	Saved value
axel_mean_pe	50.40323
binomial_mean	30.0355
binomial_variance	21.1201
dark_mean	0.54458
dark_variance	0.5446029
electronics_noise_mean_pe	-0.0009798307
electronics_noise_rms_pe	0.5003612
float32_charge_storage_relative_error	8.516565e-11
stationary_inverse_mean_pe	80.57573
stationary_inverse_rms_pe	11.31597
stationary_mean_pe	50.39924
transient_inverse_mean_pe	95.3881
transient_mean_pe	56.05311
two_trigger_mean	1.577185
two_trigger_quadrature	1.579483
two_trigger_variance	0.07313562